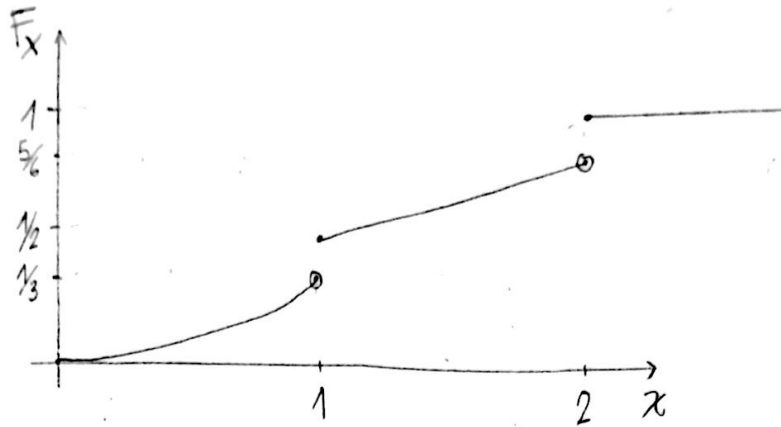


$$3.1) F_X(x) = \frac{x^3}{3} \mathbb{1}_{\{0 \leq x < 1\}} + \frac{2x+1}{6} \mathbb{1}_{\{1 \leq x < 2\}} + \mathbb{1}_{\{x \geq 2\}}$$



$$a) E[X] = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + \int_0^1 x^3 dx + \int_1^2 \frac{2x}{6} dx = \frac{5}{4}$$

$$b) E[X|X < 1] = \frac{E[X \mathbb{1}_{\{X < 1\}}]}{P(X < 1)} = \frac{\int_0^1 x^3 dx}{\int_0^1 x^2 dx} = \frac{3}{4}$$

$$E[X|X \leq 1] = \frac{\int_0^1 x^3 dx + 1 \cdot \frac{1}{6}}{\frac{1}{3}} = \frac{5}{4}$$

$$3.2) E[X] = (-2) \cdot \frac{1}{3} + 2 \cdot \frac{1}{3} + \int_{-1}^1 \frac{1}{6} x dx = 0$$

a) ↗

$$b) E[X| |X| = 2] = \frac{-2 \cdot \frac{1}{3} + 2 \cdot \frac{1}{3}}{\frac{1}{3} + \frac{1}{3}} = 0$$

$$3.3) \quad a) \quad E[N_k] = \frac{k}{p} = \frac{8}{5} k$$

$$b) \quad E[N_1] = \frac{8}{5} \quad e) \quad E[N_2] = \frac{16}{5}$$

$$3.4) \quad \Omega = \{1, 2, 3, 4, 5, 6\}$$

$$\mathcal{A} = \{\emptyset, \{1, 2\}, \{3, 4\}, \{5, 6\}, \{1, 2, 3, 4\}, \{1, 2, 5, 6\}, \{3, 4, 5, 6\}, \Omega\}$$

$$P(\{1, 2\}) = \frac{1}{6} \quad P(\{3, 4\}) = \frac{1}{3} \quad P(\{5, 6\}) = \frac{1}{2}$$

$$a) \quad \text{Definir } X: \Omega \rightarrow \mathbb{R} / \{-1, \frac{1}{2}\} \subset X(\Omega) \text{ y } E[X] = 0$$

$$P(\{1, 2, 3, 4\}) = \frac{1}{2} \quad P(\{1, 2, 5, 6\}) = \frac{2}{3} \quad P(\{3, 4, 5, 6\}) = \frac{5}{6}$$

OPCIÓN 1

$$X(1) = X(2) = -1$$

$$X(3) = X(4) = \frac{1}{2}$$

$$X(5) = X(6) = 0$$

$$\Rightarrow E[X] = 0$$

y hay más opciones...

$$X(3) = X(4) = -1$$

$$X(1) = X(2) = X(5) = X(6) = \frac{1}{2}$$

$$\Rightarrow E[X] = 0$$

$$b) E[X|X > -1] = \frac{E[X \mathbb{1}_{\{X > -1\}}]}{P(X > -1)} = \frac{1/2}{P(X=0, 1/2)}$$

opción 1 ↗

$$= \frac{1/2}{1/2 + 1/3} = \frac{3}{5}$$

$$c) \{-1, 1\} \subset X(\Omega)$$

OPCIÓN 1

NO ES ÚNICA

$$X(5) = X(6) = 1$$

$$X(1) = X(2) = X(3) = X(4) = -1$$

$$\Rightarrow E[X] = 0$$

$$E[X|X > -1] = \frac{1/2}{1/2} = 1$$

opción 1 ↗

$$3.5) E[N] = 10$$

$$P_N(1) = \frac{10^1}{1!}$$

$$E[N|N \leq 4] = \frac{E[N \cdot 1_{\{N \leq 4\}}]}{P(N \leq 4)}$$

$$= \frac{1 \cdot 10e^{-10} + \frac{2 \cdot 10^2 e^{-10}}{2} + \frac{3 \cdot 10^3 e^{-10}}{6} + \frac{4 \cdot 10^4 e^{-10}}{24}}{\frac{10^0 e^{-10}}{1} + \frac{10^1 e^{-10}}{1} + \frac{10^2 e^{-10}}{2} + \frac{10^3 e^{-10}}{6} + \frac{10^4 e^{-10}}{24}}$$

$$\approx 3,533$$

$$3.6) E[T] = \frac{1}{\lambda} = 3$$

$$E[T|T \leq 2] = \frac{\int_0^2 t \cdot \frac{1}{3} e^{-t/3} dt}{1 - e^{-2/3}} = 0,8897$$

$$\underbrace{E[T]}_3 = \underbrace{E[T|T \leq 2]}_{0,8897} \cdot \underbrace{P(T \leq 2)}_{1 - e^{-2/3}} + \underbrace{E[T|T > 2]}_{2+3} \cdot \underbrace{P(T > 2)}_{e^{-2/3}}$$

$$\Rightarrow E[T|T \leq 2] = 0,8897$$

$$7) a) F_X(x) = \frac{9!}{3!5!} x^3 (1-x)^5 \mathbb{1}_{\{0 < x < 1\}}$$

$$= 9 \cdot \binom{8}{5} x^3 (1-x)^5 \mathbb{1}_{\{0 < x < 1\}}$$

$$E[X] = \int_0^1 \frac{9!}{3!5!} x^4 (1-x)^5 dx = \frac{2}{5}$$

$$b) F_X(x) = \frac{1}{6} x^3 e^{-x} \mathbb{1}_{\{x > 0\}}$$

$$X \sim \text{GAMMA}(4, 1)$$

$$\Rightarrow E[X] = 4$$

$$3.8) Z \sim N(0, 1)$$

$$a) E[Z|Z > z_0] = \frac{\int_{z_0}^{+\infty} x \varphi(x) dx}{P(Z > z_0)} = \frac{\int_{z_0}^{+\infty} x \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx}{1 - \Phi(z_0)} = \frac{\frac{1}{\sqrt{2\pi}} \int_{z_0}^{+\infty} x e^{-x^2/2} dx}{1 - \Phi(z_0)} =$$

$$= \frac{\frac{1}{\sqrt{2\pi}} e^{-z_0^2/2}}{1 - \Phi(z_0)} = \frac{\varphi(z_0)}{1 - \Phi(z_0)}$$

$$b) \frac{\overbrace{z_0 \int_{z_0}^{+\infty} \varphi(x) dx}^{1 - \Phi(z_0)}}{1 - \Phi(z_0)} \leq \frac{\int_{z_0}^{+\infty} x \varphi(x) dx}{1 - \Phi(z_0)} = \frac{\varphi(z_0)}{1 - \Phi(z_0)}$$

$$\Rightarrow 1 - \Phi(z_0) \leq \frac{\varphi(z_0)}{z_0}$$

$$3.9) \quad P(\max(z_1, z_2) > 5) \leq \frac{2e^{-\frac{25}{2}}}{5\sqrt{2\pi}}$$

$$\begin{aligned} P(\max(z_1, z_2) > 5) &= P(z_1 > 5 \cup z_2 > 5) = \\ &= P(z_1 > 5) + P(z_2 > 5) - P(z_1 > 5 \cap z_2 > 5) \\ &\leq P(z_1 > 5) + P(z_2 > 5) = 1 - \Phi(5) + 1 - \Phi(5) \\ &\leq \frac{\varphi(5)}{5} + \frac{\varphi(5)}{5} = \frac{2}{5} \frac{e^{-\frac{25}{2}}}{\sqrt{2\pi}} \end{aligned}$$

continua 3.7)

$$c) \quad E[X | X > \frac{1}{2}] = \frac{E[X \mathbb{1}_{\{X > \frac{1}{2}\}}]}{P(X > \frac{1}{2})} =$$

$$= \frac{\int_{\frac{1}{2}}^{+\infty} \frac{1}{6} x^4 e^{-x} dx}{\int_{\frac{1}{2}}^{+\infty} \frac{1}{6} x^3 e^{-x} dx} =$$

10) $L = \text{"longitud del alambre"} \sim \text{EXP}(1/60)$

a)

$$A(L) = \frac{L^2}{4\pi}$$

$$2\pi r = L \Rightarrow r = \frac{L}{2\pi} \Rightarrow A = \pi \left(\frac{L}{2\pi}\right)^2 = \frac{L^2}{4\pi}$$

$$\begin{aligned} E[A] &= E\left[\frac{L^2}{4\pi}\right] = \frac{1}{4\pi} E[L^2] = \frac{1}{4\pi} (V(L) + E^2[L]) \\ &= \frac{1}{4\pi} (3600 + 3600) = 572,96 \end{aligned}$$

b) $A = \text{"área del círculo"} \sim \text{EXP}(1/15)$

$$P(A) = 2\sqrt{\pi A}$$

$$A = \pi r^2 \Rightarrow r = \sqrt{\frac{A}{\pi}} \Rightarrow 2\pi \sqrt{\frac{A}{\pi}} = L \Rightarrow L = 2\sqrt{\pi} \sqrt{A}$$

$$\begin{aligned} E[P] &= E[2\sqrt{\pi A}] = 2\sqrt{\pi} E[\sqrt{A}] = 2\sqrt{\pi} \underbrace{\int_0^{+\infty} \sqrt{a} e^{-a/15} da}_{\cong 182,5} \end{aligned}$$

$$3.12) F_X(x) = F_Y(y) = \frac{1}{\pi} \Rightarrow F_{XY}(x,y) = \frac{1}{\pi^2}$$

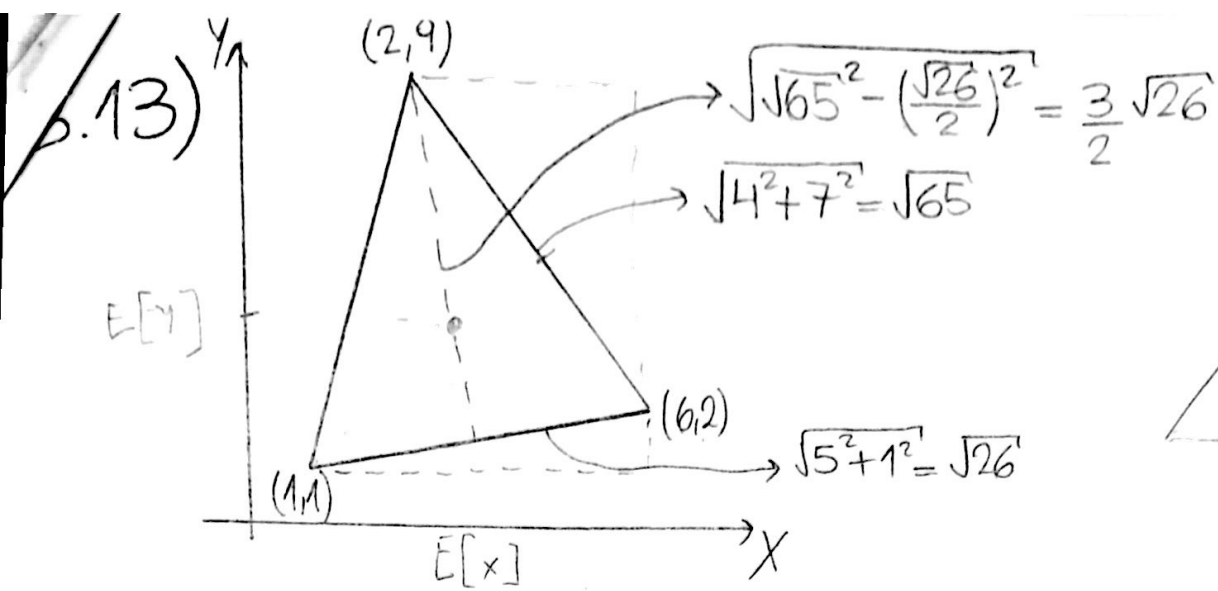
$$\begin{aligned} E[X \sin(XY)] &= \int_0^\pi \int_0^\pi x \sin(xy) \frac{1}{\pi^2} dy dx = \\ &= \int_0^\pi \frac{1}{\pi^2} x \left[\frac{\cos(xy)}{x} \right]_0^\pi dx = -\frac{1}{\pi^2} \int_0^\pi (\cos(\pi x) - 1) dx = \\ &= -\frac{1}{\pi^2} \left(\frac{\sin(\pi x)}{\pi} - x \right)_0^\pi = -\frac{1}{\pi^2} \left(\frac{\sin(\pi^2)}{\pi} - \pi \right) = \\ &= \frac{1}{\pi} - \frac{\sin(\pi^2)}{\pi^3} \end{aligned}$$

$$3.11) T^* = \min(T, 1)$$

$$E[T^*] = \int_0^1 t^* e^{-t^*} dt^* + 1 \cdot e^{-1} = 0,632$$

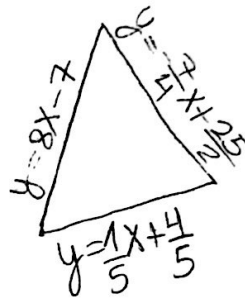
$$V(T^*) = E[T^{*2}] - E^2[T^*]$$

$$= \int_0^1 t^{*2} e^{-t^*} dt^* + 1 \cdot e^{-1} - 0,632^2 = 0,129$$



$$\Rightarrow \text{AREA} = \frac{\sqrt{26} \times \frac{3}{2} \sqrt{26}}{2} = 19,5 = \frac{39}{2}$$

$$F_{xy}(x,y) = \frac{2}{39} \mathbb{1}_{\{(x,y) \in A\}}$$



$$E[X] = \iint_A x F_{xy} dx dy =$$

$$= \int_1^2 \int_{\frac{1}{5}x + \frac{4}{5}}^{8x-7} x \cdot \frac{2}{39} dy dx + \int_2^6 \int_{\frac{1}{5}x + \frac{4}{5}}^{-\frac{1}{4}x + \frac{25}{2}} x \cdot \frac{2}{39} dy dx =$$

$$= \frac{2}{39} \int_1^2 \left(\frac{39}{5} x^2 - \frac{39}{5} x \right) dx + \frac{2}{39} \int_2^6 \left(-\frac{39}{20} x^2 + \frac{117}{10} x \right) dx = 3$$

$$E[Y] = \iint_A y F_{xy} dx dy = \int_1^2 \int_{\frac{1}{5}x + \frac{4}{5}}^{8x-7} y \frac{2}{39} dy dx + \int_2^6 \int_{\frac{1}{5}x + \frac{4}{5}}^{-\frac{1}{4}x + \frac{25}{2}} y \frac{2}{39} dy dx = 4$$

$$3.15) X \sim U(8, 10) \quad E[X] = 9 \quad V(X) = \frac{1}{3}$$

$$a) E[Y] = E[2(X-1)] = 2E[X] - 2 = 16$$

$$V[Y] = V[2(X-1)] = 4V(X) = \frac{4}{3}$$

$$b) E[Y] = E[2X^2 + 1] = 2E[X^2] + 1 = \\ = 2(V(X) + E^2[X]) + 1 = \frac{491}{3}$$

$$(V(X) + E^2[X])^2$$



$$V(Y) = V(2X^2 + 1) = 4V(X^2) = 4(E[X^4] - E^2[X^2]) = \\ = 4 \left(\int_8^{10} \frac{1}{2} x^4 dx - \frac{59536}{9} \right) = \frac{19456}{45}$$

$$c) E[Y] = E[2(X-1)(X-3)] = 2E[X^2 - 4X + 3] = \\ = 2(E[X^2] - 4E[X] + 3) = \frac{290}{3}$$

$$V(Y) = 4V(X^2 - 4X + 3) = 4(V(X^2) - 16V(X)) = \\ = 4(E[X^4] - E^2[X^2] - 16V(X)) = \frac{18496}{45}$$

$$\begin{aligned}
 d) \min_{c \in \mathbb{R}} E[(X-c)^2] &= \min_c E[X^2 - 2cX + c^2] = \\
 &= \min_c (E[X^2] - 2cE[X] + c^2) = \\
 &= \min_c \left(\frac{244}{3} - 18c + c^2 \right) \Rightarrow 2c - 18 = 0 \Rightarrow c = 9 \\
 &\Rightarrow \min_{c \in \mathbb{R}} E[(X-c)^2] = \frac{244}{3} - 18 \cdot 9 + 9^2 = -\frac{809}{13}
 \end{aligned}$$

$$e) E[aX+b] = 0, V(aX+b) = 1$$

$$\begin{array}{l|l}
 aE[X] + b = 0 & a^2 V(X) = 1 \\
 9a + b = 0 & \frac{1}{3} a^2 = 1
 \end{array}$$

$$9\sqrt{3} + b = 0 \leftarrow a = \sqrt{3}$$

$$b = -9\sqrt{3}$$

$$316) V \sim N(6, 1) \quad E[V] = 6 \quad V(V) = 1$$

$$\begin{aligned}
 E[W] &= E[3V^2] = 3E[V^2] = 3(V(V) + E^2[V]) = \\
 &= 3(1 + 6^2) = 111
 \end{aligned}$$

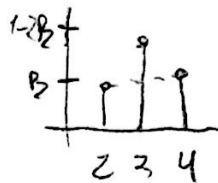
$$3.14) E[X] = 3$$

$$p_2 + p_3 + p_4 = 1 \rightarrow p_4 = 1 - p_2 - p_3$$

$$2p_2 + 3p_3 + 4p_4 = 3 = 2p_2 + 3p_3 + 4 - 4p_2 - 4p_3 = 4 - 2p_2 - p_3 = 3$$

$$1 = 2p_2 + p_3 \Rightarrow p_3 = 1 - 2p_2$$

$$p_4 = 1 - p_2 - 1 + 2p_2 = p_2$$



$$a) \arg \max_{p_2} V[X] = \arg \max_{p_2} E[X^2] - 9 = \arg \max_{p_2} E[X^2]$$

$$4p_2 + 9p_3 + 16p_4$$

11

MAX $\boxed{p_2 = 1/2}$

MIN $\boxed{p_2 = 0}$

$$\Rightarrow p_2 = 1/2$$

$$p_3 = 0$$

$$p_4 = 1/2$$

$$b) p_2 = 0$$

$$p_3 = 1$$

$$p_4 = 0$$

$$A) X_1 \sim \text{BER}(p_1) \quad X_2 \sim \text{BER}(p_2)$$

$$a) \text{COV}(X_1, X_2) = 0 = E[X_1 X_2] - E[X_1] \cdot E[X_2]$$

$$\Rightarrow E[X_1 X_2] = E[X_1] \cdot E[X_2]$$

$\Rightarrow$ son independientes

$$b) p_1 = p_2 = 0,7 = p \quad E[X_1] = 0,7 \quad E[X_2] = 0,7$$

$$\text{COV}(X_1, X_2) = 0,1 \quad V(X_1) = V(X_2) = 0,21 = p(1-p)$$

hallar $\rho(Y_1, Y_2)$

datos

$$Y_1 = X_1(1 - X_2) \quad Y_2 = X_2(1 - X_1)$$

$$\rho(Y_1, Y_2) = \frac{\text{COV}(Y_1, Y_2)}{\sigma_{Y_1} \times \sigma_{Y_2}} = \frac{E[Y_1 Y_2] - E[Y_1]E[Y_2]}{\sqrt{E[Y_1^2] - E^2[Y_1]} \sqrt{E[Y_2^2] - E^2[Y_2]}}$$

$$E[Y_1] = E[X_1(1 - X_2)] = E[X_1] - E[X_1 X_2]$$

$$E[Y_2] = E[X_2(1 - X_1)] = E[X_2] - E[X_2 X_1]$$

3.18) 3 bolas en 3 urnas C_1, C_2, C_3

X_i : "cant de bolas en c_i "

N : "cant de urnas que contienen alguna bola"

a) $E[N]$, $V(N)$, $\text{cov}(N, X_1)$

$$p_N(n) = \begin{cases} n=1 \Rightarrow X_1=3 \cup X_2=3 \cup X_3=3 \\ n=2 \Rightarrow X_1=2, X_2=1 \cup X_1=2, X_3=1 \cup \dots \\ n=3 \Rightarrow X_1=1, X_2=1, X_3=1 \end{cases}$$

$$\begin{aligned} p_N(1) &= P(X_1=3 \cup X_2=3 \cup X_3=3) = \\ &= P(X_1=3) + P(X_2=3) + P(X_3=3) = \\ &= \frac{1^3}{3^3} + \frac{1^3}{3^3} + \frac{1^3}{3^3} = \frac{1}{9} \end{aligned}$$

$$p_N(2) = \frac{\binom{3}{1}\binom{3}{2}\binom{2}{1}\binom{1}{1}}{3^3} = \frac{2}{3}$$

$$p_N(3) = \frac{\binom{3}{1}\binom{2}{1}\binom{1}{1}}{3^3} = \frac{2}{9}$$

$$E[N] = 1 \times \frac{1}{9} + 2 \times \frac{2}{9} + 3 \times \frac{2}{9} = \frac{19}{9}$$

$$V(N) = E[N^2] - E^2[N]$$

$$= \left(1 \times \frac{1}{9} + 4 \times \frac{2}{9} + 9 \times \frac{2}{9} \right) - \frac{361}{81} = \frac{26}{81}$$

$$\text{COV}(N, X_1) = E[NX_1] - E[N] \cdot E[X_1]$$

$$= \sum_{n, x_1 \in A} n x_1 p(n, x_1) - \frac{19}{9} \times \left(\sum_{x_1 \in A} x_1 p_{X_1}(x_1) \right)$$

$$p_{X_1}(x_1) = \begin{cases} x_1=0 \rightarrow \frac{2^3}{3^3} = \frac{8}{27} \\ x_1=1 \rightarrow \frac{\binom{3}{1} 2^2}{3^3} = \frac{4}{9} \\ x_1=2 \rightarrow \frac{\binom{3}{2} 2^1}{3^3} = \frac{2}{9} \\ x_1=3 \rightarrow \frac{1^3}{3^3} = \frac{1}{27} \end{cases}$$

$$E[X_1] = 1 \times \frac{4}{9} + 2 \times \frac{2}{9} + 3 \times \frac{1}{27} = \frac{11}{9}$$

$$p_{NX_1}(n, x_1) = \begin{cases} n=1, x_1=0 \rightarrow \frac{2}{27} \\ n=1, x_1=3 \rightarrow \frac{1}{27} \\ n=2, x_1=0 \rightarrow \frac{6}{27} \\ n=2, x_1=1 \rightarrow \frac{6}{27} \\ n=2, x_1=2 \rightarrow \frac{6}{27} \\ n=3, x_1=1 \rightarrow \frac{6}{27} \end{cases} \Rightarrow E[NX_1] = \frac{19}{9}$$

$$\Rightarrow \text{COV}(N, X_1) = \frac{19}{9} - \frac{19}{9} \times \frac{11}{9} = -\frac{38}{81}$$

b) para que sean independientes

$$\underbrace{p_N(1)}_{1/9} \times \underbrace{p_{X_1}(1)}_{4/9} = \underbrace{p_{NX_1}(1,1)}_0$$

$\Rightarrow$ no son independientes

c) $\text{COV}(X_i, X_j), 1 \leq i \leq j \leq 3$

calculo $p_{X_2}(X_2), p_{X_3}(X_3), p_{X_1X_2}(X_1, X_2), p_{X_1X_3}(X_1, X_3)$
y $p_{X_2X_3}(X_2, X_3)$ con sus esperanzas

$$E[X_1] = E[X_2] = E[X_3] = \frac{11}{9} \quad E[X_1^2] = E[X_2^2] = E[X_3^2] = \frac{7}{3}$$

$$p_{X_1X_2}(X_1, X_2) = \begin{cases} X_1=0, X_2=0 \rightarrow 1/27 \\ X_1=0, X_2=1 \rightarrow 3/27 \\ X_1=0, X_2=2 \rightarrow 3/27 \\ X_1=0, X_2=3 \rightarrow 1/27 \\ X_1=1, X_2=0 \rightarrow 3/27 \\ X_1=1, X_2=1 \rightarrow 6/27 \\ X_1=1, X_2=2 \rightarrow 3/27 \\ X_1=2, X_2=0 \rightarrow 3/27 \\ X_1=2, X_2=1 \rightarrow 3/27 \\ X_1=3, X_2=0 \rightarrow 1/27 \end{cases}$$

$$E[X_1X_2] = \frac{2}{3}$$

$$E[X_1X_3] = \frac{2}{3}$$

$$E[X_2X_3] = \frac{2}{3}$$

$$\text{COV}(X_1, X_1) = V(X_1) = E[X_1^2] - E^2[X_1] = \frac{7}{3} - \frac{121}{81} = \frac{68}{81}$$

$$\text{COV}(X_1, X_2) = E[X_1 X_2] - E[X_1]E[X_2] = \frac{2}{3} - \frac{11}{9} \cdot \frac{11}{9} = -\frac{67}{81}$$

$$\text{COV}(X_1, X_3) = E[X_1 X_3] - E[X_1]E[X_3] = -\frac{67}{81}$$

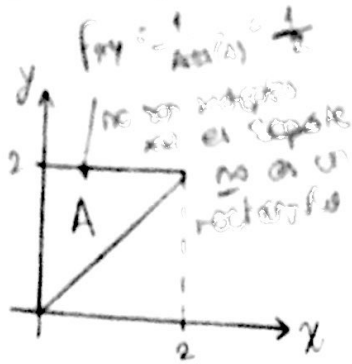
$$\text{COV}(X_2, X_2) = V(X_2) = \frac{68}{81}$$

$$\text{COV}(X_2, X_3) = E[X_2 X_3] - E[X_2]E[X_3] = -\frac{67}{81}$$

$$\text{COV}(X_3, X_3) = V(X_3) = \frac{68}{81}$$

- 3.19) a) $\text{COV}(X, Y) < 0$
 b) $\text{COV}(X, Y) = 0$
 c) $\text{COV}(X, Y) > 0$

3.20) $(X, Y) \sim U(A)$



a) $\text{COV}(X, Y) = E[XY] - E[X]E[Y] =$

$$= \int_0^2 \int_0^y xy \cdot \frac{1}{2} dx dy - \int_0^2 \int_0^y \frac{x}{2} dx dy \times \int_0^2 \int_0^y \frac{y}{2} dx dy$$

$$= 1 - \frac{2}{3} \times \frac{4}{3} = \frac{1}{9}$$

ver guia 1.2) dois caminhos

b) $\text{VAR}(X+Y) = E[(X+Y)^2] - E^2[X+Y] =$

$$= E[X^2 + 2XY + Y^2] - E[X+Y] \times E[X+Y] =$$

$$= E[X^2] + E[2XY] + E[Y^2] - E[X+Y] \times E[X+Y] =$$

$$= \int_0^2 \int_0^y \frac{x^2}{2} dx dy + 2 \int_0^2 \int_0^y xy \cdot \frac{1}{2} dx dy + \int_0^2 \int_0^y \frac{y^2}{2} dx dy - \left[\int_0^2 \int_0^y \frac{(x+y)}{2} dx dy \right]^2 =$$

$$= \frac{2}{3} + 2 \times 1 + 2 - \left(\frac{2}{3} + \frac{4}{3} \right)^2 = \frac{2}{3}$$

c) $\text{COV}(3X - Y + 2, X + Y) = \text{COV}(3X, X) + \text{COV}(3X, Y) - \text{COV}(Y, X) -$
 $-\text{COV}(Y, Y) + \text{COV}(2, X) + \text{COV}(2, Y) =$

$$= 3\text{COV}(X, X) + 3\text{COV}(X, Y) - \text{COV}(X, Y) - \text{COV}(Y, Y) =$$

$$= 3V(X) + 2\text{COV}(X, Y) - V(Y) =$$

$$= 3(E[X^2] - E^2[X]) + 2 \times \frac{1}{9} - (E[Y^2] - E^2[Y]) =$$

$$= 3 \left[\frac{2}{3} - \left(\frac{2}{3} \right)^2 \right] + \frac{2}{9} - \left[2 - \left(\frac{4}{3} \right)^2 \right] = \frac{2}{3}$$

X : "tiempo de reacción de Juan" $\sim U(0,2)$
 Y : "tiempo de reacción de María" $\sim U(0,2)$

W : "tiempo de reacción del ganador" $= \min(X, Y) = \begin{cases} X & \text{si } X < Y \\ Y & \text{si } Y < X \end{cases}$

L : "tiempo de reacción del perdedor" $= \max(X, Y) = \begin{cases} X & \text{si } Y < X \\ Y & \text{si } X < Y \end{cases}$

$$a) \hat{L} = \frac{\text{COV}(L, W)}{V(W)} (W - E[W]) + L[E[L]]$$

$$E[W] = \int_0^2 \int_0^x \frac{x}{4} dy dx + \int_0^2 \int_0^y \frac{y}{4} dy dx = \frac{1}{3} + \frac{1}{3} = \frac{2}{3}$$

$$E[L] = \int_0^2 \int_x^2 \frac{y}{4} dy dx + \int_0^2 \int_0^x \frac{x}{4} dy dx = \frac{2}{3} + \frac{2}{3} = \frac{4}{3}$$

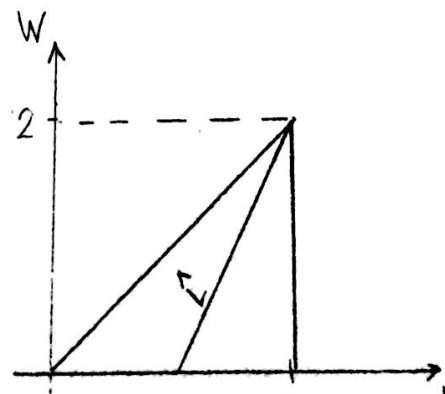
$$V(W) = E[W^2] - E^2[W] = \frac{1}{2} - \left(\frac{2}{3}\right)^2 = \frac{1}{18}$$

$$E[W^2] = \int_0^2 \int_0^x \frac{x^2}{4} dy dx + \int_0^2 \int_0^y \frac{y^2}{4} dy dx = \frac{1}{3} + \frac{1}{6} = \frac{1}{2}$$

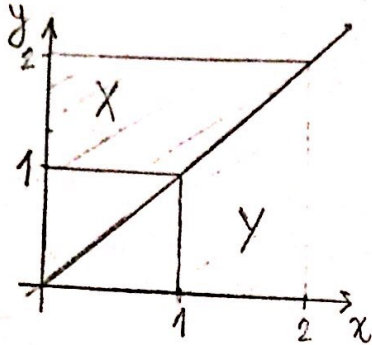
$$\text{COV}(L, W) = E[LW] - E[L]E[W] = 1 - \frac{4}{3} \times \frac{2}{3} = \frac{1}{9}$$

$$E[LW] = \int_0^2 \int_0^x \frac{xy}{4} dy dx = 1$$

$$\hat{L} = \frac{\frac{1}{9}}{\frac{1}{18}} (W - \frac{2}{3}) + \frac{4}{3} = \frac{1}{2} W + 1$$



$$b) E[W|L>1] = \frac{E[W1\{L>1\}]}{P(L>1)} =$$



$$= \frac{\int_0^2 \int_0^y \frac{x}{4} dx dy + \int_1^2 \int_0^x \frac{y}{4} dy dx}{3/4} =$$

$$= \frac{7/24 + 7/24}{3/4} = \frac{7}{9}$$

3.22)

$$f_{xy}(x,y) = \frac{5}{8\pi} e^{-\frac{25}{32}(x^2 - \frac{6}{5}xy + y^2)}$$

(RESUELTO CON
CONCEPTOS DE
GUIA 5)

$$= \frac{1}{\pi \cdot \frac{8}{5}} e^{-\frac{25}{32}((y - \frac{3}{5}x)^2 + \frac{16}{25}x^2)} =$$

$$= \frac{1}{\sqrt{\pi} \frac{8}{5}} e^{-\frac{(y - \frac{3}{5}x)^2}{32/25}} \frac{1}{\sqrt{\pi}} e^{-\frac{1}{2}x^2} = \underbrace{\frac{1}{\sqrt{2\pi} \frac{4}{5}} e^{-\frac{1}{2} \frac{(y - \frac{3}{5}x)^2}{16/25}}}_{f_{Y|X=x} = \frac{f_{xy}}{f_x}} \times \underbrace{\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}}_{f_x}$$

$$f_{Y|X=x} = \frac{f_{xy}}{f_x}$$

$$x \sim N_{est}(0,1)$$

$$Y|X=x \sim N(\frac{3}{5}x, \frac{16}{25})$$

$$E[Y|X] = \frac{3}{5}X = \psi(X) \quad \text{tiene pnte de recta}$$

5) $E[X] = 15$

demostrar $P(X \geq 60) \leq 0,25$

por Markov $\rightarrow P(X > a) \leq \frac{E[X]}{a}$

$$\Rightarrow P(X \geq 60) \leq \frac{15}{60} = 0,25$$

3.24) $E[X] = 10$, $V(X) = 15$

demostrar $P(5 < X < 15) \geq 0,4$

por Tchebycheff $\rightarrow P(|X - E[X]| \geq \varepsilon) \leq \frac{V(X)}{\varepsilon^2}$

$$\Rightarrow P(|X - 10| \geq \varepsilon) \leq \frac{15}{\varepsilon^2}$$

$$1 - P(|X - 10| < \varepsilon) \leq \frac{15}{\varepsilon^2}$$

$$1 - P(|X - 10| < 5) \leq \frac{3}{5}$$

$$P(|X - 10| < 5) \geq \frac{2}{5}$$

$$\Rightarrow P(5 < X < 15) \geq 0,4$$

3.25) X "peso de paquetes de café" $\sim N(500, \sigma^2)$

$$X_1, X_2, \dots, X_n \approx X$$

① hallar $\sigma / P(490 < \bar{X} < 510) = 0,99$

$$\bar{X} = \frac{\sum_{i=1}^{100} X_i}{100} \quad E[\bar{X}] = \frac{1}{100} \sum_{i=1}^{100} E[X_i] = 500$$

$$\underbrace{\bar{X}}_{\text{peso promedio de 100 paquetes}} \quad V(\bar{X}) = \frac{V(\sum X_i)}{100} = \frac{1}{100^2} \sum_{i=1}^{100} V(X_i) = \frac{V(X_i)}{100} = \frac{\sigma^2}{100}$$

estándares

$$\Rightarrow P(490 < \bar{X} < 510) \stackrel{!}{=} P\left(\frac{490-500}{\sigma/\sqrt{100}} < Z < \frac{510-500}{\sigma/\sqrt{100}}\right) = 0,99$$

$$P\left(-\frac{1000}{\sigma^2} < Z < \frac{1000}{\sigma^2}\right) \approx 2\Phi\left(\frac{1000}{\sigma^2}\right) - 1 \approx 0,99$$

(ESTO ES GUÍA 8
PERO NO HAY QUE
USAR EL =)

(ver (2))

$$2\Phi\left(\frac{100}{\sigma^2}\right) \approx 1,99$$

$$\Phi\left(\frac{100}{\sigma^2}\right) \approx 0,995$$

$$\frac{100}{\sigma^2} \approx \Phi^{-1}(0,995) \approx 2,5758$$

$$\Rightarrow [\sigma \approx 6,23]$$

CHEVUSHEV
hallar σ / $P(490 < \bar{X} < 510) \geq 0,99$

$$E[\bar{X}] = 500 \quad V[\bar{X}] = \frac{\sigma^2}{100}$$

$$P(|\bar{X} - 500| < 10) \geq 0,99$$

$$1 - P(|\bar{X} - 500| \geq 10) \geq 0,99$$

$$\begin{cases} P(|\bar{X} - 500| \geq 10) \leq 0,01 \end{cases}$$

$$\begin{cases} P(|\bar{X} - 500| \geq 10) \leq \frac{\sigma^2/100}{100} \Rightarrow \frac{\sigma^2}{10000} \leq 0,01 \end{cases}$$

$$\Rightarrow [\sigma \leq 10]$$

3.26) $P(|X - E[X]| < 0,01) \geq 0,95$ $X_i \sim \text{Ber}(p)$

$$X_i = \begin{cases} 0 & \text{si la persona } i \text{ no fuma} \\ 1 & \text{si la persona } i \text{ fuma} \end{cases}$$

promedio de funciones

$$X = \frac{\sum_{i=1}^n X_i}{n}$$

$$E[X] = p$$

$$V(X) = \frac{p(1-p)}{n}$$

CHEVUSHEV

$$\begin{cases} P(|X - p| < 0,01) \geq 0,95 \\ P(|X - p| \geq 0,01) \leq 0,05 \geq \frac{p(1-p)}{n(0,01)^2} \end{cases}$$

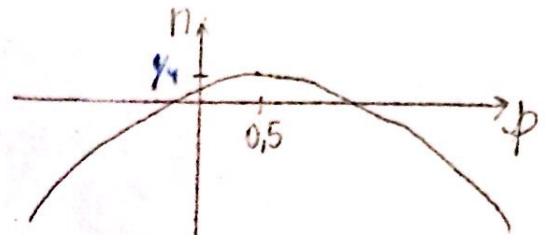
$$0,05 \geq \frac{p(1-p)}{n(0,01)^2}$$

$$n \geq \frac{p(1-p)}{0,05 \times 0,01^2}$$

⇒ derivo e igualo a cero

$$\left[\frac{p(1-p)}{0,05 \times 0,01^2} \right]' = \frac{1-2p}{0,05 \times 0,01^2} = 0 \Rightarrow p = \frac{1}{2}$$

$$n \geq \frac{\frac{1}{2}(1-\frac{1}{2})}{0,05 \times 0,01^2} \Rightarrow \boxed{n \geq 50000}$$



con asegurar que
n sea mayor al
máximo, aseguro que
sea mayor a todo p

$VC(x) = \frac{p(1-p)}{n}$ → analizando la función se ve que $0 \leq p \leq 1$
y en este intervalo la función alcanza
un max en $p = \frac{1}{2}$ independiente de n